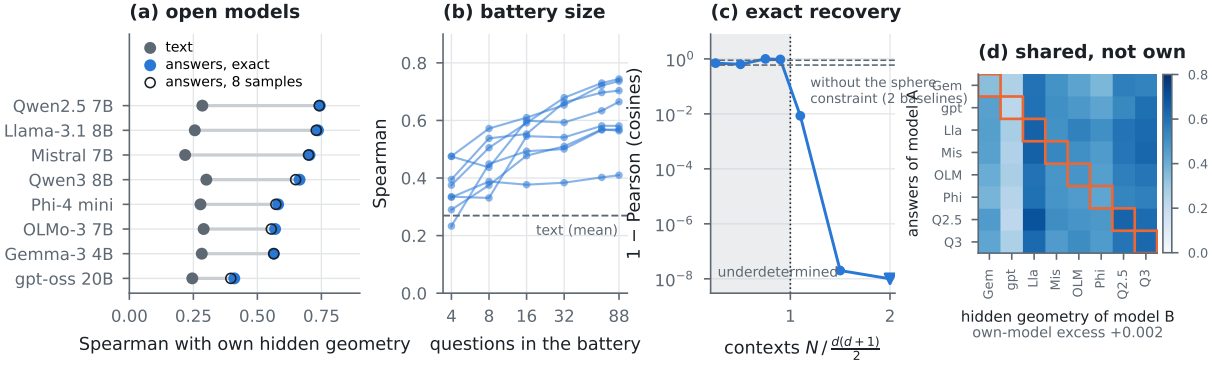
(a) open models
text
answers, exact
answers, 8 samples
Qwen2.5 7B
Llama-3.1 8B
Mistral 7B
Qwen3 8B
Phi-4 mini
OLMo-3 7B
Gemma-3 4B
gpt-oss 20B
0.00
0.25
0.50
0.75
Spearman with own hidden geometry
(b) battery size
Spearman
0.0
0.2
0.4
0.6
0.8
text (mean)
4
8
16
32
88
questions in the battery
(c) exact recovery
1 − Pearson (cosines)
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
without the sphere constraint (2 baselines)
underdetermined
1
2
contexts N / $\frac{d(d+1)}{2}$
(d) shared, not own
answers of model A
Gem
gpt
Lla
Mis
OLM
Phi
Q2.5
Q3
0.0
0.2
0.4
0.6
0.8
hidden geometry of model B
own-model excess +0.002